

Dear Editors,

Please consider the manuscript “*Typed material-flow ledgers for componentwise sustainability diagnostics*” for publication in the *Journal of Industrial Ecology*.

The manuscript develops a typed stock–flow ledger in which material types, conversion routes, conserved moieties, support-pool drawdown and service readouts are distinguished formally. It establishes the componentwise limitation of compensatory aggregation and classifies groundwater anomaly persistence, phosphate reserve-life arithmetic and fisheries removals-only pressure times as distinct descriptive quantities rather than interchangeable depletion forecasts. The accompanying supplementary material provides the theorem inventory, delay-dynamics interface, application tables and operational weak- and strong-sustainability definitions.

The manuscript is part of a linked set of three papers:

1. **Main article:** “*Typed material-flow ledgers for componentwise sustainability diagnostics*.” Public record: 10.6084/m9.figshare.33942451.
2. **Methods companion:** Abaee, A. (2026), “*Certifying a Typed Ledger: The Predicates, the Programmes, the Vintages, and the Reproduction Bundle*.” 10.6084/m9.figshare.33942469.
3. **Accounting-standards companion:** Abaee, A. (2026), “*What the Accounts Settle, and What They Leave Open: Depletion as a Cost of Production, and the Missing Step from a Rate to a Horizon*.” 10.6084/m9.figshare.33942487.

The related papers have distinct scopes: the submitted manuscript presents the industrial-ecology framework and findings; the methods companion specifies the certification procedures and reproduction protocol; and the accounting-standards companion examines the distinction between recording depletion flows and inferring horizons.

The author declares no competing interests. The work received no funding.

Thank you for your consideration.

Sincerely,

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